Grad-CAM: Why did you say that?
Visual Explanations from Deep Networks via Gradient-based Localization

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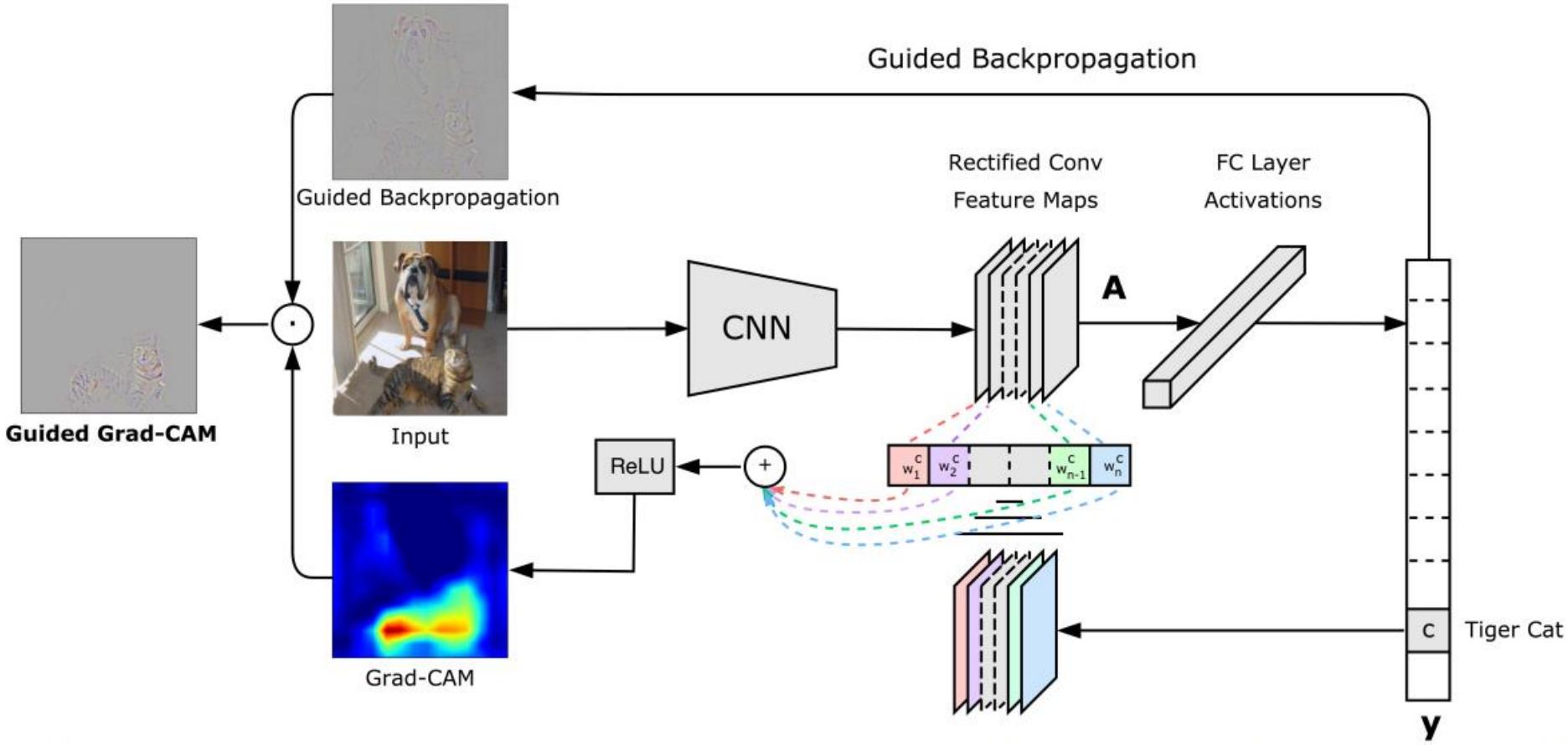
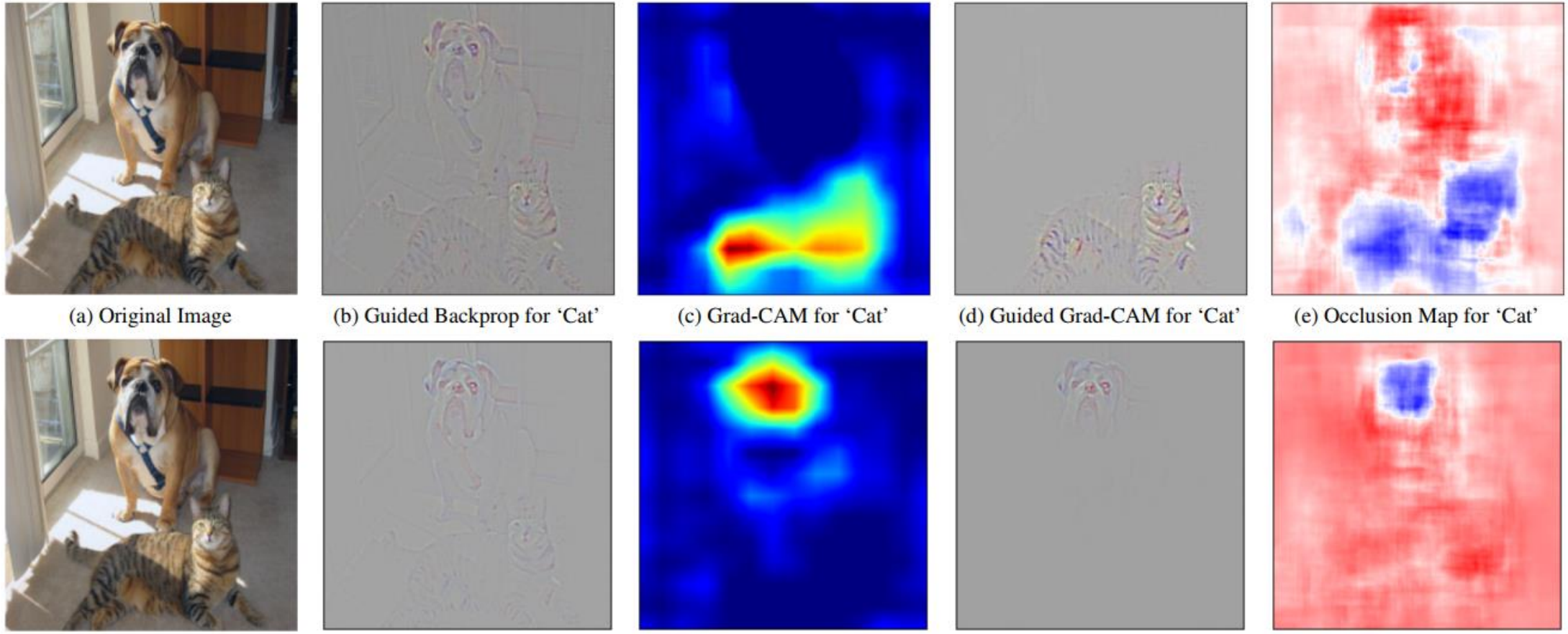


Figure 2: Grad-CAM overview: Given an image, and a category ('tiger cat') as input, we foward propagate the image through the model to obtain the raw class scores before softmax. The gradients are set to zero for all classes except the desired class (tiger cat), which is set to 1. This signal is then backpropagated to the rectified convolutional feature map of interest, where we can compute the coarse Grad-CAM localization (blue heatmap). Finally, we pointwise multiply the heatmap with guided backpropagation to get Guided Grad-CAM visualizations which are both high-resolution and class-discriminative.



(a) Original Image (b) Guided Backprop for 'Cat' (c) Grad-CAM for 'Cat' (d) Guided Grad-CAM for 'Cat' (e) Occlusion Map for 'Cat'

(f) Original Image (g) Guided Backprop for 'Dog' (h) Grad-CAM for 'Dog' (i) Guided Grad-CAM for 'Dog' (j) Occlusion Map for 'Dog'

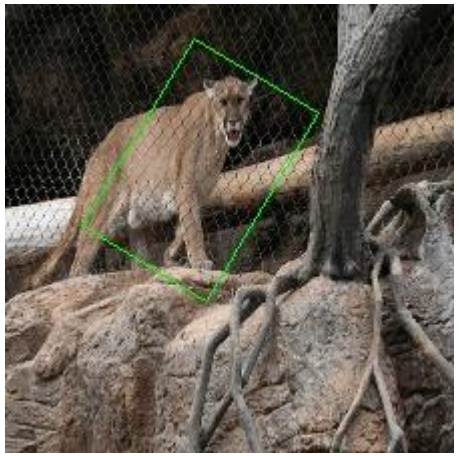
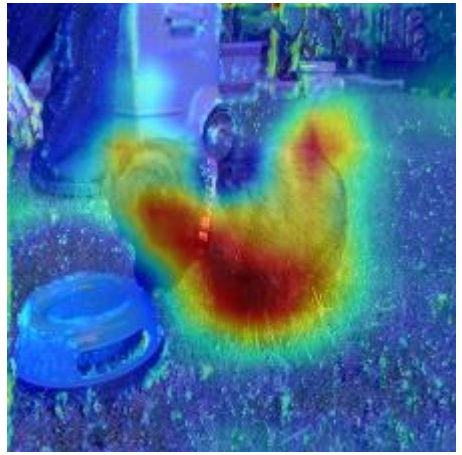
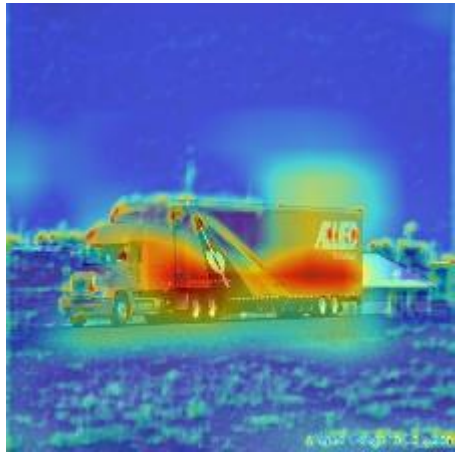
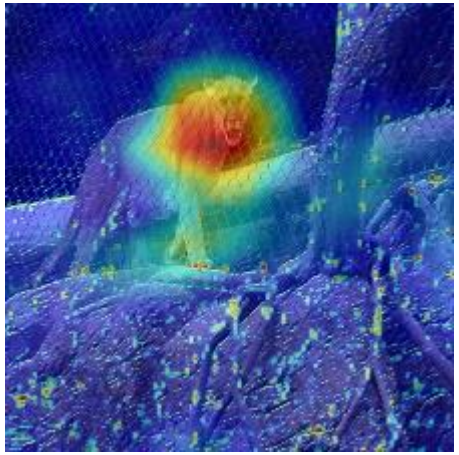



Figure 1. (a) Original image with a cat and a dog. (b-e) Support for the cat category according to various visualizations. (b) Guided Backpropagation [38]: provides high-resolution visualization of contributing features, (c) Grad-CAM (Ours): localizes class-discriminative regions, (d) Combining (b) and (c) gives Guided Grad-CAM, which gives high-resolution visualizations that are class-discriminative. Interestingly, the localizations achieved by our Grad-CAM technique (c) are very similar to results from occlusion sensitivity (e), while being much cheaper to compute. Note that in (e), blue corresponds to evidence *for* the class while in (c) blue indicates regions with low score for the class. Figure best viewed in color.